

DSC680 Ayden Trusty Week 10 Milestone 2 Notebook

May 31, 2026

0.1 Game Size vs. Success

0.2 Analyzing Video Game File Sizes, Sales, and Reviews Over Time

Project Goal: Explore whether larger video games tend to sell more and receive better review scores. Are they worth the space they take up on your hard drive?

```
[20]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import requests
import time
import re

# Optional later
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

pd.set_option("display.max_columns", None)
```

```
[9]: # Kaggle datasets
steam_df = pd.read_csv("games.csv")
sales_df = pd.read_csv("Video_Games_Sales_as_at_22_Dec_2016.csv")

steam_df.head()
```

```
[9]:
```

	AppID	Name	Release date	\
2539430	Black Dragon Mage Playtest	Aug 1, 2023	0 - 0	
496350	Supipara - Chapter 1 Spring Has Come!	Jul 29, 2016	0 - 20000	
1034400	Mystery Solitaire The Black Raven	May 6, 2019	0 - 20000	
3292190	- Vtuber Paranoia	Oct 31, 2024	0 - 20000	
3631080	Maze Quest VR	Apr 24, 2025	0 - 20000	

	Estimated owners	Peak CCU	Required age	Price	Discount	DLC count	\
2539430	0	0	0.00	0		0	
496350	0	0	5.24	65		0	
1034400	0	0	4.99	0		0	
3292190	1	0	8.99	0		1	

3631080	0	0	4.99	0	0
---------	---	---	------	---	---

			About the game \
2539430			NaN

496350	Springtime, April: when the cherry trees come ...
1034400	Immerse yourself in the most beloved, mystical...
3292190	synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...
3631080	Its not just a Maze; its a Quest! Enter the ca...

	Supported languages	Full audio languages	\
2539430		[]	[]
496350		['English']	[]
1034400	['English', 'French', 'German', 'Russian']		[]
3292190		['Korean']	['Korean']
3631080		['English']	['English']

	Reviews		Header image	\
2539430	NaN	https://shared.akamai.steamstatic.com/store_it...		
496350	NaN	https://shared.akamai.steamstatic.com/store_it...		
1034400	NaN	https://shared.akamai.steamstatic.com/store_it...		
3292190	NaN	https://shared.akamai.steamstatic.com/store_it...		
3631080	NaN	https://shared.akamai.steamstatic.com/store_it...		

	Website	\
2539430	NaN	
496350	http://mangagamer.org/supipara	
1034400	https://www.facebook.com/8FloorGames/	
3292190	NaN	
3631080	https://www.realityexpanded.com/books-games	

	Support url	Support email	\
2539430	NaN	NaN	
496350	http://mangagamer.com	support@mangagamer.com	
1034400	https://www.facebook.com/8FloorGames	support@8floor.net	
3292190	NaN	yujingamesc@gmail.com	
3631080	https://www.realityexpanded.com	support@realityexpanded.com	

	Windows	Mac	Linux	Metacritic score	Metacritic url	User score	\
2539430	True	False	False	0	NaN	0	
496350	True	False	False	0	NaN	0	
1034400	True	True	False	0	NaN	0	
3292190	True	False	False	0	NaN	0	
3631080	True	False	False	0	NaN	0	

	Positive	Negative	Score rank	Achievements	Recommendations	\
2539430	0	0	NaN	0	0	
496350	252	3	NaN	0	231	

1034400	21	3	NaN	0	0
3292190	0	0	NaN	19	0
3631080	0	0	NaN	0	0

	Notes \
2539430	NaN
496350	NaN
1034400	NaN
3292190	The game includes the following elements. 1. G...
3631080	NaN

	Average playtime forever	Average playtime two weeks \
2539430	0	0
496350	8	0
1034400	0	0
3292190	0	0
3631080	0	0

	Median playtime forever	Median playtime two weeks \
2539430	0	0
496350	8	0
1034400	0	0
3292190	0	0
3631080	0	0

	Developers	Publishers \
2539430	NaN	NaN
496350	minori	MangaGamer
1034400	Somer Games	8floor
3292190		
3631080	Reality Expanded LLC	Reality Expanded LLC

	Categories \
2539430	NaN
496350	Single-player,Steam Trading Cards,Steam Cloud,...
1034400	Single-player,Family Sharing
3292190	Single-player,Steam Achievements,Family Sharing
3631080	Single-player,VR Only,Steam Leaderboards,Famil...

	Genres \
2539430	NaN
496350	Adventure
1034400	Casual
3292190	Casual,Indie,Simulation
3631080	Action,Early Access

Tags \

2539430		NaN
496350	Adventure, Visual Novel, Anime, Cute	
1034400	Casual, Card Game, Solitaire, Puzzle, Hidden Objec...	
3292190		NaN
3631080		NaN

	Screenshots	Movies
2539430	https://shared.akamai.steamstatic.com/store_it...	NaN
496350	https://shared.akamai.steamstatic.com/store_it...	NaN
1034400	https://shared.akamai.steamstatic.com/store_it...	NaN
3292190	https://shared.akamai.steamstatic.com/store_it...	NaN
3631080	https://shared.akamai.steamstatic.com/store_it...	NaN

```
[5]: sales_df.head()
```

```
[5]:
```

	Name	Platform	Year_of_Release	Genre	Publisher	\
0	Wii Sports	Wii	2006.0	Sports	Nintendo	
1	Super Mario Bros.	NES	1985.0	Platform	Nintendo	
2	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	
3	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	
4	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	

	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales	Critic_Score	\
0	41.36	28.96	3.77	8.45	82.53	76.0	
1	29.08	3.58	6.81	0.77	40.24	NaN	
2	15.68	12.76	3.79	3.29	35.52	82.0	
3	15.61	10.93	3.28	2.95	32.77	80.0	
4	11.27	8.89	10.22	1.00	31.37	NaN	

	Critic_Count	User_Score	User_Count	Developer	Rating
0	51.0	8	322.0	Nintendo	E
1	NaN	NaN	NaN	NaN	NaN
2	73.0	8.3	709.0	Nintendo	E
3	73.0	8	192.0	Nintendo	E
4	NaN	NaN	NaN	NaN	NaN

```
[4]: steam_df.columns
```

```
[4]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
        'Required age', 'Price', 'DiscountDLC count', 'About the game',
        'Supported languages', 'Full audio languages', 'Reviews',
        'Header image', 'Website', 'Support url', 'Support email', 'Windows',
        'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
        'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
        'Notes', 'Average playtime forever', 'Average playtime two weeks',
        'Median playtime forever', 'Median playtime two weeks', 'Developers',
        'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies'],
        dtype=object)
```

```
dtype='object')
```

```
[6]: sales_df.columns
```

```
[6]: Index(['Name', 'Platform', 'Year_of_Release', 'Genre', 'Publisher', 'NA_Sales',  
        'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales', 'Critic_Score',  
        'Critic_Count', 'User_Score', 'User_Count', 'Developer', 'Rating'],  
        dtype='object')
```

```
[11]: steam_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
Index: 122611 entries, 2539430 to 4141790  
Data columns (total 39 columns):  
#   Column                                Non-Null Count  Dtype  
---  -  
0   AppID                                122610 non-null  object  
1   Name                                 122611 non-null  object  
2   Release date                         122611 non-null  object  
3   Estimated owners                     122611 non-null  int64  
4   Peak CCU                             122611 non-null  int64  
5   Required age                         122611 non-null  float64  
6   Price                                122611 non-null  int64  
7   DiscountDLC count                    122611 non-null  int64  
8   About the game                       114162 non-null  object  
9   Supported languages                  122611 non-null  object  
10  Full audio languages                 122611 non-null  object  
11  Reviews                             12070 non-null   object  
12  Header image                         122530 non-null  object  
13  Website                              49676 non-null   object  
14  Support url                          54142 non-null   object  
15  Support email                        100348 non-null  object  
16  Windows                              122611 non-null  bool  
17  Mac                                  122611 non-null  bool  
18  Linux                                122611 non-null  bool  
19  Metacritic score                     122611 non-null  int64  
20  Metacritic url                       4256 non-null    object  
21  User score                           122611 non-null  int64  
22  Positive                             122611 non-null  int64  
23  Negative                             122611 non-null  int64  
24  Score rank                           40 non-null      float64  
25  Achievements                         122611 non-null  int64  
26  Recommendations                      122611 non-null  int64  
27  Notes                                22458 non-null   object  
28  Average playtime forever              122611 non-null  int64  
29  Average playtime two weeks            122611 non-null  int64  
30  Median playtime forever                122611 non-null  int64  
31  Median playtime two weeks              122611 non-null  int64
```

```

32 Developers          114174 non-null object
33 Publishers          113702 non-null object
34 Categories          113658 non-null object
35 Genres              114198 non-null object
36 Tags                83346 non-null object
37 Screenshots         116593 non-null object
38 Movies              0 non-null float64
dtypes: bool(3), float64(3), int64(14), object(19)
memory usage: 35.0+ MB

```

```
[12]: sales_df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16719 entries, 0 to 16718
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Name                  16717 non-null  object
1   Platform              16719 non-null  object
2   Year_of_Release       16450 non-null  float64
3   Genre                 16717 non-null  object
4   Publisher             16665 non-null  object
5   NA_Sales              16719 non-null  float64
6   EU_Sales              16719 non-null  float64
7   JP_Sales              16719 non-null  float64
8   Other_Sales           16719 non-null  float64
9   Global_Sales          16719 non-null  float64
10  Critic_Score          8137 non-null   float64
11  Critic_Count          8137 non-null   float64
12  User_Score            10015 non-null  object
13  User_Count            7590 non-null   float64
14  Developer             10096 non-null  object
15  Rating                9950 non-null   object
dtypes: float64(9), object(7)
memory usage: 2.0+ MB

```

```
[13]: steam_df.duplicated().sum()
```

```
[13]: 7
```

```
[15]: steam_df = steam_df.drop_duplicates()
```

```
[16]: steam_df.duplicated().sum()
```

```
[16]: 0
```

```
[14]: sales_df.duplicated().sum()
```

[14]: 0

Now that I've made sure that both files imported successfully, I've just been inspecting them to see if there are duplicates and removing those. Then I'll move on to NAN and make sure everything is clean before analysis.

```
[10]: # Test Steam API Data Collection
appid = 1245620 # Elden Ring

url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

response = requests.get(url)
data = response.json()

data
```

```
[10]: {'1245620': {'success': True,
  'data': {'type': 'game',
    'name': 'ELDEN RING',
    'steam_appid': 1245620,
    'required_age': '16',
    'is_free': False,
    'controller_support': 'full',
    'dlc': [2778590, 2778580],
    'detailed_description': '<h1>ELDEN RING Shadow of the Erdtree
Edition</h1><p><span class="bb_img_ctn"><video class="bb_img" autoplay muted
loop playsinline poster="https://shared.akamai.steamstatic.com/store_item_assets
/steam/apps/1245620/extras/6e68c7bbe71819c202f190579f8d4f12.poster.avif?t=176788
3716" width=616 height=346 ><source src="https://shared.akamai.steamstatic.com/s
tore_item_assets/steam/apps/1245620/extras/6e68c7bbe71819c202f190579f8d4f12.webm
?t=1767883716" type="video/webm; codecs=vp9"><source src="https://shared.akamai.
steamstatic.com/store_item_assets/steam/apps/1245620/extras/6e68c7bbe71819c202f1
90579f8d4f12.mp4?t=1767883716" type="video/mp4"></video></span><br>ELDEN RING
Shadow of the Erdtree Edition includes:<br><ul class="bb_ul"><li>ELDEN
RING<br></li><li>ELDEN RING Shadow of the Erdtree
expansion</li></ul></p><br><h1>ELDEN RING Shadow of the Erdtree Deluxe
Edition</h1><p><span class="bb_img_ctn"><video class="bb_img" autoplay muted
loop playsinline poster="https://shared.akamai.steamstatic.com/store_item_assets
/steam/apps/1245620/extras/0856a96484c4fb0a35eee2db28320da2.poster.avif?t=176788
3716" width=616 height=346 ><source src="https://shared.akamai.steamstatic.com/s
tore_item_assets/steam/apps/1245620/extras/0856a96484c4fb0a35eee2db28320da2.webm
?t=1767883716" type="video/webm; codecs=vp9"><source src="https://shared.akamai.
steamstatic.com/store_item_assets/steam/apps/1245620/extras/0856a96484c4fb0a35ee
e2db28320da2.mp4?t=1767883716" type="video/mp4"></video></span><br>ELDEN RING
Shadow of the Erdtree Deluxe Edition includes:<br><ul class="bb_ul"><li>ELDEN
RING<br></li><li>ELDEN RING Shadow of the Erdtree expansion<br></li><li>ELDEN
RING Digital Artbook & Original Soundtrack<br></li><li>ELDEN RING Shadow of
the Erdtree Artbook & Soundtrack</li></ul></p><br><h1>About the
```

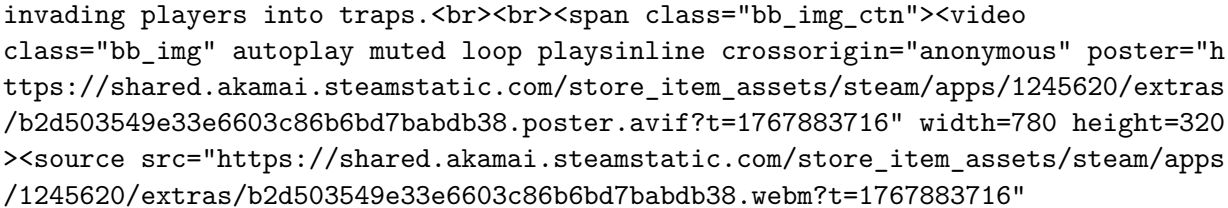
Game

THE CRITICALLY ACCLAIMED FANTASY ACTION RPG

Rise, Tarnished, and be guided by grace to brandish the power of the Elden Ring.

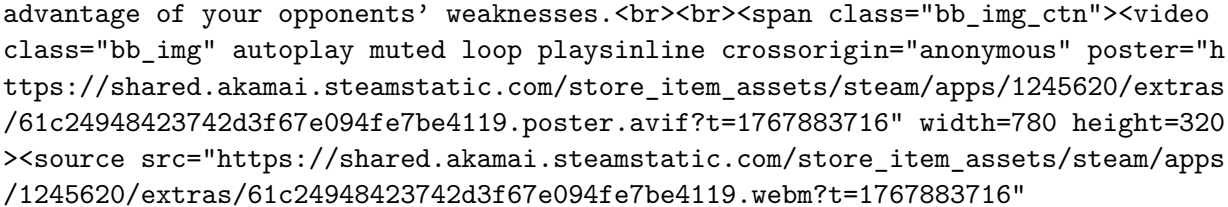
• A Breathtaking World Full of Excitement and Mystery

The Lands Between are part of a vast continent where magnificent open fields and huge dungeons with complex and three-dimensional designs are seamlessly connected. As you explore, the joy of discovering unknown and overwhelming threats awaits you. Mastery of the terrain and knowledge of its secrets can help you overcome enemies and defeat formidable bosses or lead invading players into traps.



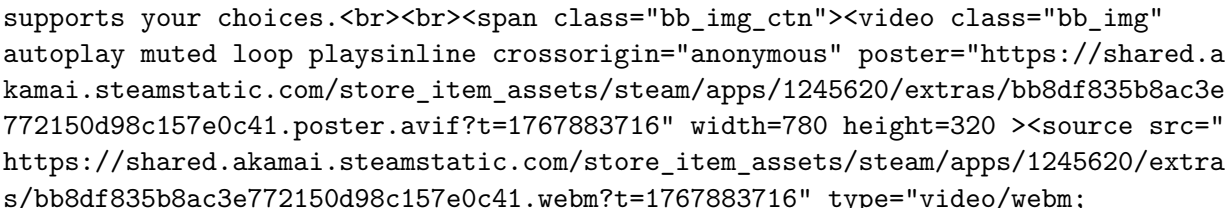
• Defeat Challenging Foes in Intense Combat

Combat in ELDEN RING is simple to learn yet offers hidden depths of mastery. As you seek to become the Elden Lord, you'll need to explore the balance between attacking and avoiding damage, use a wide variety of weapons, spells, & summons, and perfect your timing to take advantage of your opponents' weaknesses.



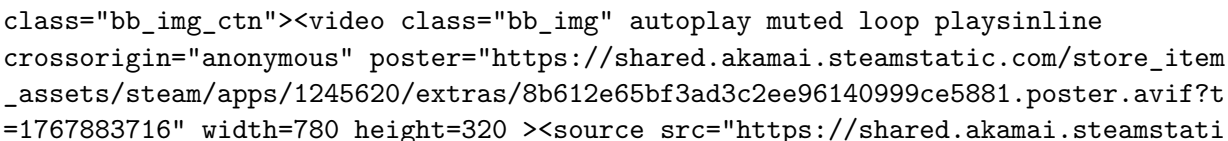
• Create and Build Your Own Character

In addition to customizing the appearance of your character, there are countless ways to combine the weapons, armor, usable items, and magic that you equip. You can develop your character according to your play style. No matter if you prefer bold physical confrontation, tactical spellcasting, or the subtle art of stealth, you'll be able to find gear that supports your choices.



• An Epic Drama Born from a Myth Created by George R.R. Martin

The founding mythology of Elden Ring was written by George R. R. Martin and adapted into a rich multilayered story. Intersecting goals and desires between the characters create an intense narrative that weaves throughout the Lands Between. The events of the game can unravel in many ways, depending on your interventions.



c.com/store_item_assets/steam/apps/1245620/extras/8b612e65bf3ad3c2ee96140999ce5881.webm?t=1767883716" type="video/webm; codecs=vp9"></video><h2 class="bb_tag" >• Play Alongside a Massive Worldwide Community</h2>The Tarnished community is massive and active. Your friends may already be among them. You can play with up to two other Tarnished as your cooperative teammates, either by inviting them using a shared password or by summoning them from a pool of nearby community members.
There are also ample opportunities to face off against other players, either through co-op invasions, invited duels, or the many player battle options available in the three Colosseums.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8bed670f075526277b71e13a1c614f51.poster.avif?t=1767883716" width=780 height=320 ><source src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8bed670f075526277b71e13a1c614f51.webm?t=1767883716" type="video/webm; codecs=vp9"></video>',

'about_the_game': 'THE CRITICALLY ACCLAIMED FANTASY ACTION RPG
<i>Rise, Tarnished, and be guided by grace to brandish the power of the Elden Ring.</i><h2 class="bb_tag" >• A Breathtaking World Full of Excitement and Mystery</h2>The Lands Between are part of a vast continent where magnificent open fields and huge dungeons with complex and three-dimensional designs are seamlessly connected. As you explore, the joy of discovering unknown and overwhelming threats awaits you.
Mastery of the terrain and knowledge of its secrets can help you overcome enemies and defeat formidable bosses or lead invading players into traps.

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No matter if you prefer bold physical confrontation, tactical spellcasting, or the subtle art of stealth, you'll be able to find gear that supports your choices.

<video class="bb_img"

autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/bb8df835b8ac3e772150d98c157e0c41.poster.avif?t=1767883716" width=780 height=320 ><source src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/bb8df835b8ac3e772150d98c157e0c41.webm?t=1767883716" type="video/webm; codecs=vp9"></video><h2 class="bb_tag" >• An Epic Drama Born from a Myth Created by George R.R. Martin</h2>The founding mythology of Elden Ring was written by George R. R. Martin and adapted into a rich multilayered story. Intersecting goals and desires between the characters create an intense narrative that weaves throughout the Lands Between. The events of the game can unravel in many ways, depending on your interventions.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8b612e65bf3ad3c2ee96140999ce5881.poster.avif?t=1767883716" width=780 height=320 ><source src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8b612e65bf3ad3c2ee96140999ce5881.webm?t=1767883716" type="video/webm; codecs=vp9"></video><h2 class="bb_tag" >• Play Alongside a Massive Worldwide Community</h2>The Tarnished community is massive and active. Your friends may already be among them. You can play with up to two other Tarnished as your cooperative teammates, either by inviting them using a shared password or by summoning them from a pool of nearby community members.
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'short_description': 'THE CRITICALLY ACCLAIMED FANTASY ACTION RPG. Rise, Tarnished, and be guided by grace to brandish the power of the Elden Ring and become an Elden Lord in the Lands Between.',

'supported_languages': 'English*, French, Italian, German, Spanish - Spain, Japanese, Korean, Polish, Portuguese - Brazil, Russian, Simplified Chinese, Spanish - Latin America, Thai, Traditional Chinese, Arabic
*languages with full audio support',

'reviews': '"Put a ring on it."
10/10 - IGN

"An unmissable open-world masterpiece."
10/10 - Gaming Bible

"Exploration is jaw dropping."
5/5 - Games Radar
',

'header_image': 'https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/header.jpg?t=1767883716',

'capsule_image': 'https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/capsule_231x87.jpg?t=1767883716',

'capsule_imagev5': 'https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/capsule_184x69.jpg?t=1767883716',

'website': None,

```

    'pc_requirements': {'minimum': '<strong>Minimum:</strong><br><ul
class="bb_ul"><li>Requires a 64-bit processor and operating
system<br></li><li><strong>OS:</strong> Windows
10<br></li><li><strong>Processor:</strong> INTEL CORE I5-8400 or AMD RYZEN 3
3300X<br></li><li><strong>Memory:</strong> 12 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1060 3 GB or AMD
RADEON RX 580 4 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
space<br></li><li><strong>Sound Card:</strong> Windows Compatible Audio
Device<br></li><li><strong>Additional Notes:</strong> </li></ul>',
    'recommended': '<strong>Recommended:</strong><br><ul
class="bb_ul"><li>Requires a 64-bit processor and operating
system<br></li><li><strong>OS:</strong> Windows
10/11<br></li><li><strong>Processor:</strong> INTEL CORE I7-8700K or AMD RYZEN 5
3600X<br></li><li><strong>Memory:</strong> 16 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1070 8 GB or AMD
RADEON RX VEGA 56 8 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
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    'background': 'https://store.akamai.steamstatic.com/images/storepagebackgroun
d/app/1245620?t=1767883716',
    'background_raw': 'https://shared.akamai.steamstatic.com/store_item_assets/st
eam/apps/1245620/page_bg_raw.jpg?t=1767883716',
    'content_descriptors': {'ids': [2, 5], 'notes': None},
    'ratings': {'esrb': {'rating': 'm',
      'descriptors': 'Blood and Gore\r\nLanguage\r\nSuggestive
Themes\r\nViolence',
      'required_age': '16',
      'use_age_gate': 'true',
      'interactive_elements': 'In-Game Purchases'},
    'dejus': {'rating': '16',
      'descriptors': 'Violência\r\nCompras on-line',
      'use_age_gate': 'true',

```

```

    'required_age': '16'},
    'pegi': {'rating': '16',
    'descriptors': 'Violence\r\nIn-Game Purchases',
    'use_age_gate': 'true',
    'required_age': '16'},
    'usk': {'rating': '16', 'required_age': '16', 'use_age_gate': 'true'},
    'nzoflc': {'rating': 'r13',
    'descriptors': 'Violence & horror scenes',
    'required_age': '13',
    'use_age_gate': 'true'},
    'fpb': {'rating': '16', 'required_age': '16', 'use_age_gate': 'true'},
    'csrr': {'rating': 'R', 'use_age_gate': 'true', 'required_age': '18'},
    'cero': {'rating': 'd',
    'descriptors': ' ',
    'required_age': '17',
    'use_age_gate': 'true'},
    'crl': {'rating': '16', 'use_age_gate': 'true', 'required_age': '16'}}}}}}

```

```
[24]: sample_appid = steam_df["AppID"].iloc[0]
```

```
sample_appid
```

```
[24]: 'Black Dragon Mage Playtest'
```

```
[17]: steam_df.isnull().sum().sort_values(ascending=False)
```

```

[17]: Movies                122604
Score rank                 122564
Metacritic url            118348
Reviews                   110534
Notes                     100146
Website                    72934
Support url                68463
Tags                       39258
Support email              22256
Categories                  8953
Publishers                  8908
About the game              8449
Developers                  8432
Genres                     8413
Screenshots                 6018
Header image                 81
AppID                       1
Mac                           0
Linux                        0
Release date                 0
Estimated owners             0

```

Peak CCU	0
Required age	0
Price	0
DiscountDLC count	0
Median playtime two weeks	0
Median playtime forever	0
Average playtime two weeks	0
Average playtime forever	0
Supported languages	0
Recommendations	0
Achievements	0
Full audio languages	0
Negative	0
Positive	0
User score	0
Windows	0
Name	0
Metacritic score	0
dtype:	int64

```
[18]: sales_df.isnull().sum().sort_values(ascending=False)
```

```
[18]: User_Count      9129
      Critic_Score   8582
      Critic_Count   8582
      Rating         6769
      User_Score     6704
      Developer      6623
      Year_of_Release 269
      Publisher       54
      Name            2
      Genre           2
      Platform        0
      NA_Sales        0
      EU_Sales        0
      JP_Sales        0
      Other_Sales     0
      Global_Sales    0
      dtype: int64
```

```
[23]: steam_df.columns
```

```
[23]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
        'Required age', 'Price', 'DiscountDLC count', 'About the game',
        'Supported languages', 'Full audio languages', 'Reviews',
        'Header image', 'Website', 'Support url', 'Support email', 'Windows',
        'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
```

```
'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
'Notes', 'Average playtime forever', 'Average playtime two weeks',
'Median playtime forever', 'Median playtime two weeks', 'Developers',
'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies'],
dtype='object')
```

```
[26]: steam_df["Release date"] = pd.to_datetime(
    steam_df["Release date"],
    errors="coerce"
)

steam_df["Release year"] = (
    steam_df["Release date"].dt.year
)
```

C:\Users\ayden\AppData\Local\Temp\ipykernel_27360\3847174024.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
steam_df["Release date"] = pd.to_datetime(
```

```
[27]: def owner_midpoint(owner_range):

    try:
        low, high = owner_range.split(" - ")

        return (int(low) + int(high)) / 2

    except:
        return np.nan
```

```
[28]: steam_df["Estimated owners numeric"] = (
    steam_df["Estimated owners"]
    .apply(owner_midpoint)
)
```

```
[29]: steam_df["Review ratio"] = (
    steam_df["Positive"] /
    (steam_df["Positive"] + steam_df["Negative"])
)
```

```
[30]: sample_appids = (
    steam_df["AppID"]
    .dropna()
    .head(50)
)
```

```
[52]: def get_game_storage(appid):

    url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

    try:
        response = requests.get(url, timeout=10)
        data = response.json()

        app_data = data.get(str(appid), {})

        if not app_data.get("success"):
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        game = app_data.get("data", {})

        if game is None:
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        requirements = game.get("pc_requirements")

        storage_size = extract_storage_size(requirements)

        return {
            "AppID": appid,
            "install_size_gb": storage_size
        }

    except Exception as e:
        print("Error:", appid, e)
        return {
            "AppID": appid,
            "install_size_gb": None
        }
```

```
[47]: def extract_storage_size(requirements_text):

    if requirements_text is None:
        return None

    text = str(requirements_text)
```

```

# Specifically search for storage-related lines
storage_match = re.search(
    r'Storage.*?(\\d+)\\s?GB',
    text,
    re.IGNORECASE
)

if storage_match:
    return float(storage_match.group(1))

return None

```

```

[54]: test_appid = 1245620

url = f"https://store.steampowered.com/api/appdetails?appids={test_appid}"

response = requests.get(url)

data = response.json()

data[str(test_appid)]["success"]

```

[54]: True

```

[55]: game = data[str(test_appid)]["data"]

game.get("pc_requirements")

```

```

[55]: {'minimum': '<strong>Minimum:</strong><br><ul class="bb_ul"><li>Requires a
64-bit processor and operating system<br></li><li><strong>OS:</strong> Windows
10<br></li><li><strong>Processor:</strong> INTEL CORE I5-8400 or AMD RYZEN 3
3300X<br></li><li><strong>Memory:</strong> 12 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1060 3 GB or AMD
RADEON RX 580 4 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
space<br></li><li><strong>Sound Card:</strong> Windows Compatible Audio
Device<br></li><li><strong>Additional Notes:</strong> </li></ul>',
'recommended': '<strong>Recommended:</strong><br><ul class="bb_ul"><li>Requires
a 64-bit processor and operating system<br></li><li><strong>OS:</strong> Windows
10/11<br></li><li><strong>Processor:</strong> INTEL CORE I7-8700K or AMD RYZEN 5
3600X<br></li><li><strong>Memory:</strong> 16 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1070 8 GB or AMD
RADEON RX VEGA 56 8 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
space<br></li><li><strong>Sound Card:</strong> Windows Compatible Audio
Device<br></li><li><strong>Additional Notes:</strong> </li></ul>'}
```

```
[56]: extract_storage_size(game.get("pc_requirements"))
```

```
[56]: 60.0
```

This section of the project uses the Steam Store API to gather install size information for video games using each game's unique Steam AppID. The API returns metadata and system requirement information for each game, including storage requirements that are often embedded within text or HTML formatting. Python functions and regular expressions are used to extract the storage size values and convert them into usable numerical data. The cleaned install size data is then merged back into the main dataset containing reviews, ownership estimates, and sales-related metrics. This process allows the project to analyze relationships between game install size, commercial success, and player reception. I had a bit of troubleshooting trying to get the API return to specifically get the storage requirements. At first it pulled nothing, then a 12, which I know isn't the full download size for Elden Ring (no way at all), that must be the RAM recommendation/requirement as listed in the requirements above. It was just grabbing anything with GB attached to it at first, so I had to edit that. Now it's finally working and pulled 60 GB for install size.

```
[57]: get_game_storage(1245620)
```

```
[57]: {'AppID': 1245620, 'install_size_gb': 60.0}
```

```
[62]: def get_game_storage(appid):

    url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

    try:
        response = requests.get(url, timeout=10)
        data = response.json()

        app_data = data.get(str(appid), {})

        if not app_data.get("success"):
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        game = app_data.get("data", {})

        if game is None:
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        requirements = game.get("pc_requirements")

        storage_size = extract_storage_size(requirements)
```



```

    return {
        "AppID": appid,
        "install_size_gb": storage_size
    }

except Exception as e:
    print("Error:", appid, e)
    return {
        "AppID": appid,
        "install_size_gb": None
    }

```

```

[63]: steam_sample = (
    steam_df
    .dropna(subset=["AppID"])
    .sort_values("Positive", ascending=False)
    .head(100)
)

sample_appids = steam_sample["AppID"]

```

```

[64]: storage_results = []

for appid in sample_appids:

    result = get_game_storage(appid)

    print(result)

    storage_results.append(result)

    time.sleep(1)

```

```

Error: Counter-Strike 2 'NoneType' object has no attribute 'get'
{'AppID': 'Counter-Strike 2', 'install_size_gb': None}
Error: Dota 2 'NoneType' object has no attribute 'get'
{'AppID': 'Dota 2', 'install_size_gb': None}
Error: Grand Theft Auto V Legacy 'NoneType' object has no attribute 'get'
{'AppID': 'Grand Theft Auto V Legacy', 'install_size_gb': None}
Error: PUBG: BATTLEGROUNDS 'NoneType' object has no attribute 'get'
{'AppID': 'PUBG: BATTLEGROUNDS', 'install_size_gb': None}
Error: Terraria 'NoneType' object has no attribute 'get'
{'AppID': 'Terraria', 'install_size_gb': None}
Error: Tom Clancy's Rainbow Six® Siege X 'NoneType' object has no attribute
'get'
{'AppID': "Tom Clancy's Rainbow Six® Siege X", 'install_size_gb': None}
Error: Garry's Mod 'NoneType' object has no attribute 'get'

```

```

{'AppID': 'Garry's Mod', 'install_size_gb': None}
Error: Black Myth: Wukong 'NoneType' object has no attribute 'get'
{'AppID': 'Black Myth: Wukong', 'install_size_gb': None}
Error: Rust 'NoneType' object has no attribute 'get'
{'AppID': 'Rust', 'install_size_gb': None}
Error: Team Fortress 2 'NoneType' object has no attribute 'get'
{'AppID': 'Team Fortress 2', 'install_size_gb': None}
Error: ELDEN RING 'NoneType' object has no attribute 'get'
{'AppID': 'ELDEN RING', 'install_size_gb': None}
Error: Left 4 Dead 2 'NoneType' object has no attribute 'get'
{'AppID': 'Left 4 Dead 2', 'install_size_gb': None}
Error: Wallpaper Engine 'NoneType' object has no attribute 'get'
{'AppID': 'Wallpaper Engine', 'install_size_gb': None}
Error: Stardew Valley 'NoneType' object has no attribute 'get'
{'AppID': 'Stardew Valley', 'install_size_gb': None}
Error: Euro Truck Simulator 2 'NoneType' object has no attribute 'get'
{'AppID': 'Euro Truck Simulator 2', 'install_size_gb': None}
Error: The Witcher 3: Wild Hunt 'NoneType' object has no attribute 'get'
{'AppID': 'The Witcher 3: Wild Hunt', 'install_size_gb': None}
Error: HELLDIVERS 2 'NoneType' object has no attribute 'get'
{'AppID': 'HELLDIVERS 2', 'install_size_gb': None}
Error: Baldur's Gate 3 'NoneType' object has no attribute 'get'
{'AppID': 'Baldur's Gate 3', 'install_size_gb': None}
Error: Phasmophobia 'NoneType' object has no attribute 'get'
{'AppID': 'Phasmophobia', 'install_size_gb': None}
Error: Cyberpunk 2077 'NoneType' object has no attribute 'get'
{'AppID': 'Cyberpunk 2077', 'install_size_gb': None}
Error: Red Dead Redemption 2 'NoneType' object has no attribute 'get'
{'AppID': 'Red Dead Redemption 2', 'install_size_gb': None}
Error: Apex Legends 'NoneType' object has no attribute 'get'
{'AppID': 'Apex Legends', 'install_size_gb': None}
Error: Among Us 'NoneType' object has no attribute 'get'
{'AppID': 'Among Us', 'install_size_gb': None}
Error: Dead by Daylight 'NoneType' object has no attribute 'get'
{'AppID': 'Dead by Daylight', 'install_size_gb': None}
Error: ARK: Survival Evolved 'NoneType' object has no attribute 'get'
{'AppID': 'ARK: Survival Evolved', 'install_size_gb': None}
Error: The Forest 'NoneType' object has no attribute 'get'
{'AppID': 'The Forest', 'install_size_gb': None}
Error: PAYDAY 2 'NoneType' object has no attribute 'get'
{'AppID': 'PAYDAY 2', 'install_size_gb': None}
Error: Warframe 'NoneType' object has no attribute 'get'
{'AppID': 'Warframe', 'install_size_gb': None}
Error: Rocket League® 'NoneType' object has no attribute 'get'
{'AppID': 'Rocket League®', 'install_size_gb': None}
Error: Unturned 'NoneType' object has no attribute 'get'
{'AppID': 'Unturned', 'install_size_gb': None}
Error: Destiny 2 'NoneType' object has no attribute 'get'

```

```
{'AppID': 'Destiny 2', 'install_size_gb': None}
Error: War Thunder 'NoneType' object has no attribute 'get'
{'AppID': 'War Thunder', 'install_size_gb': None}
Error: Don't Starve Together 'NoneType' object has no attribute 'get'
{'AppID': "Don't Starve Together", 'install_size_gb': None}
Error: Lethal Company 'NoneType' object has no attribute 'get'
{'AppID': 'Lethal Company', 'install_size_gb': None}
Error: Valheim 'NoneType' object has no attribute 'get'
{'AppID': 'Valheim', 'install_size_gb': None}
Error: Dying Light 'NoneType' object has no attribute 'get'
{'AppID': 'Dying Light', 'install_size_gb': None}
Error: Monster Hunter: World 'NoneType' object has no attribute 'get'
{'AppID': 'Monster Hunter: World', 'install_size_gb': None}
Error: Portal 2 'NoneType' object has no attribute 'get'
{'AppID': 'Portal 2', 'install_size_gb': None}
```

```
-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[64], line 11
      7 print(result)
      9 storage_results.append(result)
----> 11 time.sleep(1)

KeyboardInterrupt:
```

```
[60]: storage_df = pd.DataFrame(storage_results)

storage_df.head()
```

```
[60]:
```

	AppID	install_size_gb
0	Counter-Strike 2	None
1	Dota 2	None
2	Grand Theft Auto V Legacy	None
3	PUBG: BATTLEGROUNDS	None

```
[61]: storage_df["install_size_gb"].notna().sum()
```

```
[61]: 0
```

```
[65]: steam_df[["AppID", "Name", "Positive"]].head()
```

```
[65]:
```

	AppID	Name	Positive
2539430	Black Dragon Mage Playtest	Aug 1, 2023	0
496350	Supipara - Chapter 1 Spring Has Come!	Jul 29, 2016	252
1034400	Mystery Solitaire The Black Raven	May 6, 2019	21
3292190	- Vtuber Paranoia	Oct 31, 2024	0
3631080	Maze Quest VR	Apr 24, 2025	0

```
[66]: steam_sample = (
        steam_df
        .dropna(subset=["AppID"])
        .sort_values("Positive", ascending=False)
        .head(50)
    )

    sample_appids = steam_sample["AppID"]
```

```
[67]: sample_appids.head()
```

```
[67]: 730                Counter-Strike 2
      570                Dota 2
      271590    Grand Theft Auto V Legacy
      578080    PUBG: BATTLEGROUNDS
      105600    Terraria
      Name: AppID, dtype: object
```

```
[68]: steam_df.columns
```

```
[68]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
        'Required age', 'Price', 'DiscountDLC count', 'About the game',
        'Supported languages', 'Full audio languages', 'Reviews',
        'Header image', 'Website', 'Support url', 'Support email', 'Windows',
        'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
        'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendations',
        'Notes', 'Average playtime forever', 'Average playtime two weeks',
        'Median playtime forever', 'Median playtime two weeks', 'Developers',
        'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies',
        'Release year', 'Estimated owners numeric', 'Review ratio'],
        dtype='object')
```

```
[69]: steam_df = steam_df.reset_index()
```

```
[70]: steam_df[["AppID", "Name"]].head()
```

```
[70]:
```

	AppID	Name
0	Black Dragon Mage Playtest	Aug 1, 2023
1	Supipara - Chapter 1 Spring Has Come!	Jul 29, 2016
2	Mystery Solitaire The Black Raven	May 6, 2019
3	- Vtuber Paranoia	Oct 31, 2024
4	Maze Quest VR	Apr 24, 2025

```
[71]: steam_sample = (
        steam_df
        .dropna(subset=["AppID"])
        .sort_values("Positive", ascending=False)
        .head(50)
```

```
)

sample_appids = steam_sample["AppID"]

sample_appids.head()
```

```
[71]: 45509          Counter-Strike 2
      4193          Dota 2
      59122    Grand Theft Auto V Legacy
      8878          PUBG: BATTLEGROUNDS
      10927          Terraria
      Name: AppID, dtype: object
```

```
[72]: steam_df.head()
```

```
[72]:      index          AppID      Name Release date \
0  2539430    Black Dragon Mage Playtest    Aug 1, 2023    NaT
1   496350  Supipara - Chapter 1 Spring Has Come!    Jul 29, 2016    NaT
2  1034400    Mystery Solitaire The Black Raven    May 6, 2019    NaT
3  3292190          - Vtuber Paranoia    Oct 31, 2024    NaT
4  3631080          Maze Quest VR    Apr 24, 2025    NaT

      Estimated owners  Peak CCU  Required age  Price  DiscountDLC count \
0                0        0          0.00      0          0
1                0        0          5.24     65          0
2                0        0          4.99      0          0
3                1        0          8.99      0          1
4                0        0          4.99      0          0

          About the game \
0                NaN
1  Springtime, April: when the cherry trees come ...
2  Immerse yourself in the most beloved, mystical...
3  synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...
4  Its not just a Maze; its a Quest! Enter the ca...

          Supported languages Full audio languages Reviews \
0                []                []    NaT
1                ['English']                []    NaT
2  ['English', 'French', 'German', 'Russian']                []    NaT
3                ['Korean']                ['Korean']    NaT
4                ['English']                ['English']    NaT

          Header image \
0  https://shared.akamai.steamstatic.com/store_it...
1  https://shared.akamai.steamstatic.com/store_it...
2  https://shared.akamai.steamstatic.com/store_it...
```

3 https://shared.akamai.steamstatic.com/store_it...
 4 https://shared.akamai.steamstatic.com/store_it...

	Website \
0	NaN
1	http://mangagamer.org/supipara
2	https://www.facebook.com/8FloorGames/
3	NaN
4	https://www.realityexpanded.com/books-games

	Support url	Support email	Windows \
0	NaN	NaN	True
1	http://mangagamer.com	support@mangagamer.com	True
2	https://www.facebook.com/8FloorGames	support@8floor.net	True
3	NaN	yujingamesc@gmail.com	True
4	https://www.realityexpanded.com	support@realityexpanded.com	True

	Mac	Linux	Metacritic score	Metacritic url	User score	Positive \
0	False	False	0	NaN	0	0
1	False	False	0	NaN	0	252
2	True	False	0	NaN	0	21
3	False	False	0	NaN	0	0
4	False	False	0	NaN	0	0

	Negative	Score rank	Achievements	Recommendations \
0	0	NaN	0	0
1	3	NaN	0	231
2	3	NaN	0	0
3	0	NaN	19	0
4	0	NaN	0	0

	Notes \
0	NaN
1	NaN
2	NaN
3	The game includes the following elements. 1. G...
4	NaN

	Average playtime forever	Average playtime two weeks \
0	0	0
1	8	0
2	0	0
3	0	0
4	0	0

	Median playtime forever	Median playtime two weeks	Developers \
0	0	0	NaN

1	8	0	minori
2	0	0	Somer Games
3	0	0	
4	0	0	Reality Expanded LLC

	Publishers	Categories \
0	NaN	NaN
1	MangaGamer	Single-player,Steam Trading Cards,Steam Cloud,...
2	8floor	Single-player,Family Sharing
3		Single-player,Steam Achievements,Family Sharing
4	Reality Expanded LLC	Single-player,VR Only,Steam Leaderboards,Famil...

	Genres	Tags \
0	NaN	NaN
1	Adventure	Adventure,Visual Novel,Anime,Cute
2	Casual	Casual,Card Game,Solitaire,Puzzle,Hidden Objec...
3	Casual,Indie,Simulation	NaN
4	Action,Early Access	NaN

	Screenshots	Movies	Release year \
0	https://shared.akamai.steamstatic.com/store_it...	NaN	NaN
1	https://shared.akamai.steamstatic.com/store_it...	NaN	NaN
2	https://shared.akamai.steamstatic.com/store_it...	NaN	NaN
3	https://shared.akamai.steamstatic.com/store_it...	NaN	NaN
4	https://shared.akamai.steamstatic.com/store_it...	NaN	NaN

	Estimated owners numeric	Review ratio
0	NaN	NaN
1	NaN	0.988235
2	NaN	0.875000
3	NaN	NaN
4	NaN	NaN

```
[76]: steam_df = steam_df.rename(
      columns={"Real_AppID": "index"}
    )
```

```
[78]: sample_appids = steam_sample["index"].astype(int)
```

```
[79]: steam_sample = (
      steam_df
      .dropna(subset=["index"])
      .sort_values("Positive", ascending=False)
      .head(50)
    )

sample_appids = (
```

```
steam_sample["index"]
    .astype(int)
)
```

```
[80]: sample_appids.head()
```

```
[80]: 45509      730
      4193      570
      59122    271590
      8878     578080
      10927    105600
      Name: index, dtype: int32
```

```
[81]: storage_results = []

      for appid in sample_appids:

          result = get_game_storage(appid)

          print(result)

          storage_results.append(result)

          time.sleep(1)
```

```
{'AppID': 730, 'install_size_gb': 85.0}
{'AppID': 570, 'install_size_gb': 60.0}
{'AppID': 271590, 'install_size_gb': 125.0}
{'AppID': 578080, 'install_size_gb': 40.0}
{'AppID': 105600, 'install_size_gb': None}
{'AppID': 359550, 'install_size_gb': 65.0}
{'AppID': 4000, 'install_size_gb': 10.0}
{'AppID': 2358720, 'install_size_gb': 130.0}
{'AppID': 252490, 'install_size_gb': 45.0}
{'AppID': 440, 'install_size_gb': 15.0}
{'AppID': 1245620, 'install_size_gb': 60.0}
{'AppID': 550, 'install_size_gb': 13.0}
{'AppID': 431960, 'install_size_gb': 2.0}
{'AppID': 413150, 'install_size_gb': None}
{'AppID': 227300, 'install_size_gb': None}
{'AppID': 292030, 'install_size_gb': 50.0}
{'AppID': 553850, 'install_size_gb': 135.0}
{'AppID': 1086940, 'install_size_gb': 150.0}
{'AppID': 739630, 'install_size_gb': 21.0}
{'AppID': 1091500, 'install_size_gb': 70.0}
{'AppID': 1174180, 'install_size_gb': 150.0}
{'AppID': 1172470, 'install_size_gb': 75.0}
{'AppID': 945360, 'install_size_gb': None}
```



```
{'AppID': 381210, 'install_size_gb': 50.0}
{'AppID': 346110, 'install_size_gb': 60.0}
{'AppID': 242760, 'install_size_gb': 5.0}
{'AppID': 218620, 'install_size_gb': 83.0}
{'AppID': 230410, 'install_size_gb': 75.0}
{'AppID': 252950, 'install_size_gb': 20.0}
{'AppID': 304930, 'install_size_gb': 4.0}
{'AppID': 1085660, 'install_size_gb': 105.0}
{'AppID': 236390, 'install_size_gb': 70.0}
{'AppID': 322330, 'install_size_gb': 3.0}
{'AppID': 1966720, 'install_size_gb': 4.0}
{'AppID': 892970, 'install_size_gb': 1.0}
{'AppID': 239140, 'install_size_gb': None}
{'AppID': 582010, 'install_size_gb': 52.0}
{'AppID': 620, 'install_size_gb': 8.0}
{'AppID': 1938090, 'install_size_gb': None}
{'AppID': 367520, 'install_size_gb': 9.0}
{'AppID': 322170, 'install_size_gb': None}
{'AppID': 374320, 'install_size_gb': 25.0}
{'AppID': 1097150, 'install_size_gb': 2.0}
{'AppID': 1623730, 'install_size_gb': 40.0}
{'AppID': 108600, 'install_size_gb': None}
{'AppID': 960090, 'install_size_gb': None}
{'AppID': 250900, 'install_size_gb': 8.0}
{'AppID': 548430, 'install_size_gb': 3.0}
```

```
-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[81], line 11
      7 print(result)
      9 storage_results.append(result)
----> 11 time.sleep(1)
```

KeyboardInterrupt:

```
[82]: storage_df = pd.DataFrame(storage_results)

storage_df.head()
```

```
[82]:
```

	AppID	install_size_gb
0	730	85.0
1	570	60.0
2	271590	125.0
3	578080	40.0
4	105600	NaN

```
[83]: storage_df["install_size_gb"].notna().sum()
```

```
[83]: 39
```

```
[84]: steam_storage_df = steam_df.merge(
    storage_df,
    left_on="index",
    right_on="AppID",
    how="left"
)
```

```
[85]: steam_storage_df[[
    "AppID_x",
    "install_size_gb",
    "Estimated owners numeric",
    "Review ratio",
    "Metacritic score"
]].head()
```

```
[85]:
```

	AppID_x	install_size_gb	\
0	Black Dragon Mage Playtest	NaN	
1	Supipara - Chapter 1 Spring Has Come!	NaN	
2	Mystery Solitaire The Black Raven	NaN	
3	- Vtuber Paranoia	NaN	
4	Maze Quest VR	NaN	

	Estimated owners numeric	Review ratio	Metacritic score
0	NaN	NaN	0
1	NaN	0.988235	0
2	NaN	0.875000	0
3	NaN	NaN	0
4	NaN	NaN	0

```
[86]: plt.figure(figsize=(8,5))

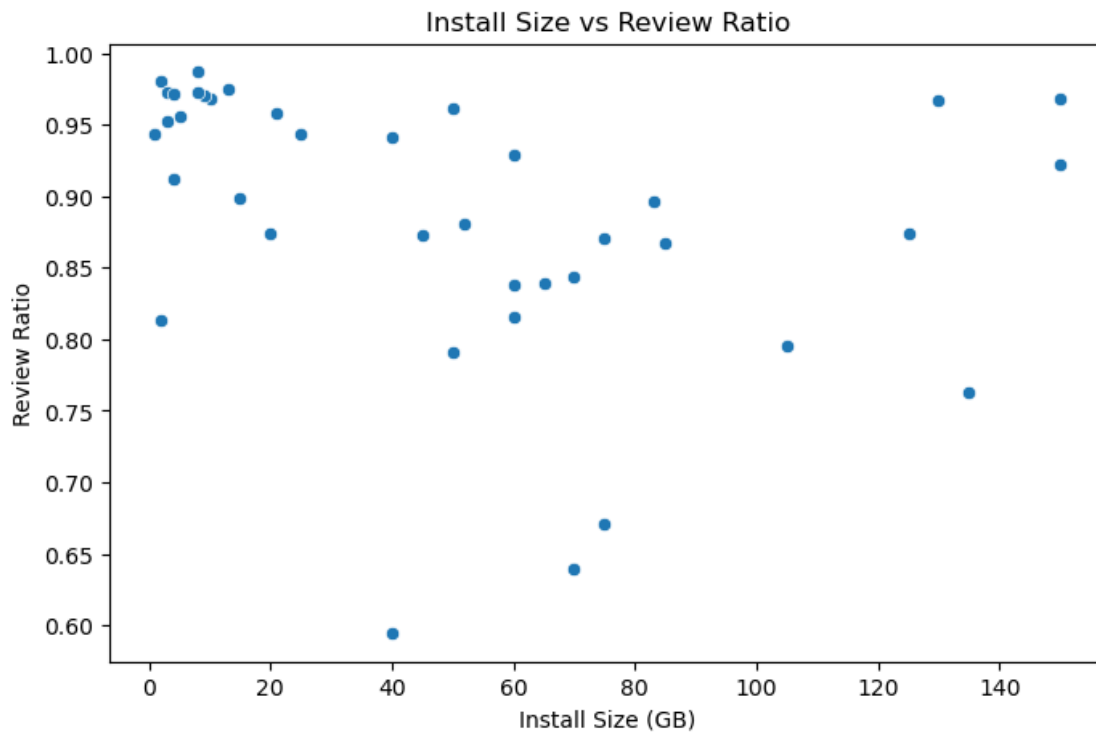
sns.scatterplot(
    data=steam_storage_df,
    x="install_size_gb",
    y="Review ratio"
)

plt.title("Install Size vs Review Ratio")

plt.xlabel("Install Size (GB)")

plt.ylabel("Review Ratio")
```

```
plt.show()
```



```
[87]: plt.figure(figsize=(8,5))

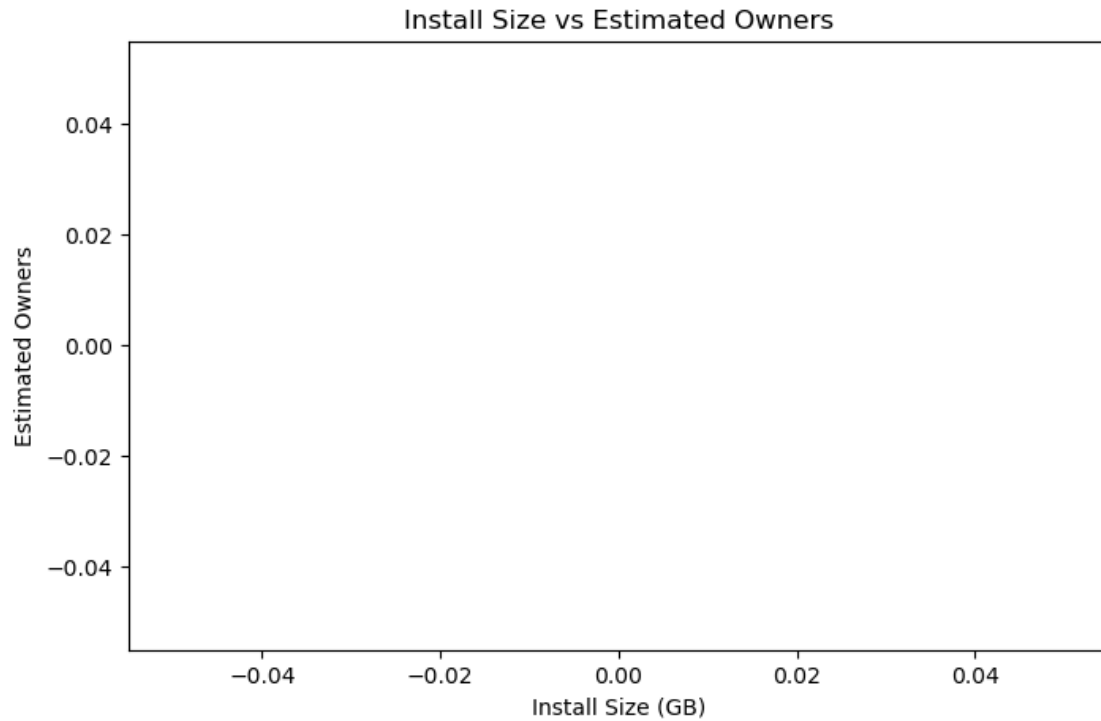
sns.scatterplot(
    data=steam_storage_df,
    x="install_size_gb",
    y="Estimated owners numeric"
)

plt.title("Install Size vs Estimated Owners")

plt.xlabel("Install Size (GB)")

plt.ylabel("Estimated Owners")

plt.show()
```



```
[92]: steam_df["Estimated owners"].head(10)
```

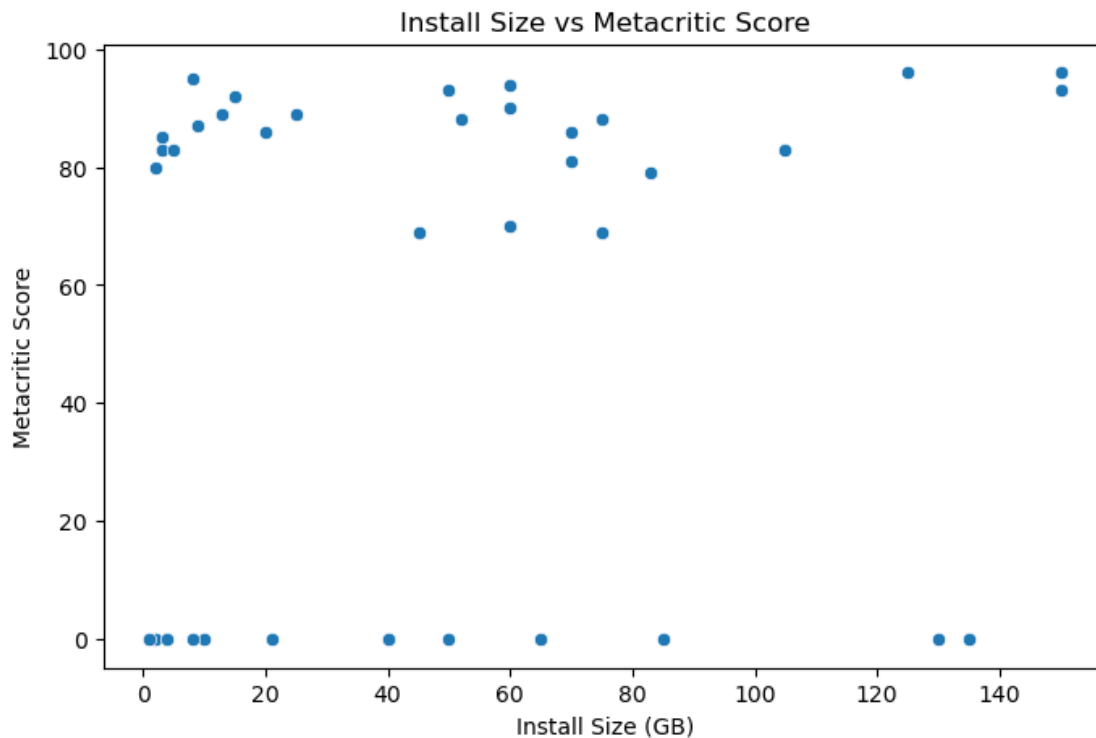
```
[92]: 0    0
      1    0
      2    0
      3    1
      4    0
      5    0
      6    8
      7    0
      8    0
      9    1
      Name: Estimated owners, dtype: int64
```

```
[88]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=steam_storage_df,
    x="install_size_gb",
    y="Metacritic score"
)

plt.title("Install Size vs Metacritic Score")
```

```
plt.xlabel("Install Size (GB)")
plt.ylabel("Metacritic Score")
plt.show()
```



```
[89]: corr_df = steam_storage_df[[
    "install_size_gb",
    "Estimated owners numeric",
    "Review ratio",
    "Metacritic score"
]].corr()

corr_df
```

```
[89]:
```

	install_size_gb	Estimated owners numeric	\
install_size_gb	1.000000		NaN
Estimated owners numeric		NaN	NaN
Review ratio	-0.302169		NaN
Metacritic score	0.135688		NaN

```

Review ratio  Metacritic score
```

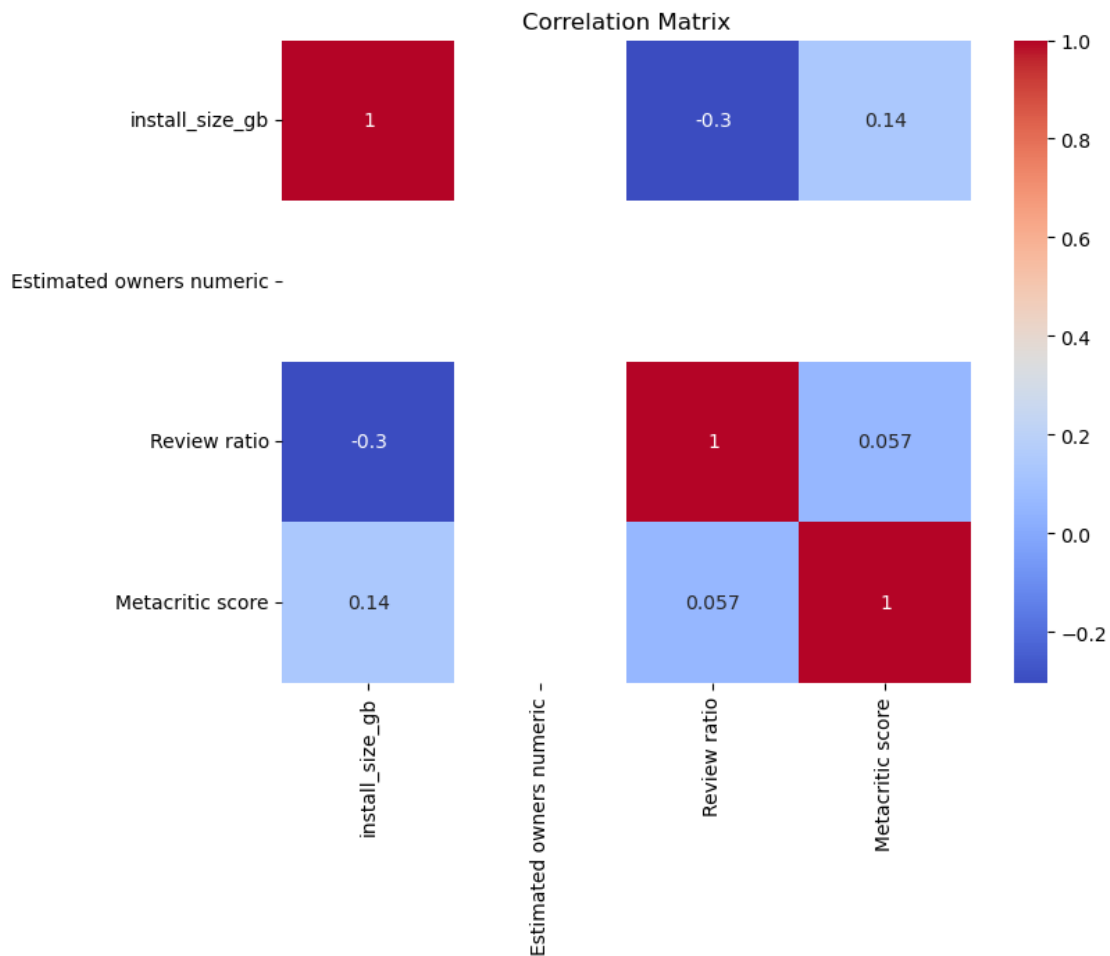
install_size_gb	-0.302169	0.135688
Estimated owners numeric	NaN	NaN
Review ratio	1.000000	0.056757
Metacritic score	0.056757	1.000000

```
[90]: plt.figure(figsize=(8,6))

sns.heatmap(
    corr_df,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```



```
[91]: largest_games = (
        steam_storage_df[[
            "AppID_x",
            "install_size_gb",
            "Estimated owners numeric"
        ]]
        .sort_values("install_size_gb", ascending=False)
        .head(10)
    )

largest_games
```

```
[91]:
```

	AppID_x	install_size_gb	Estimated owners numeric
59354	Baldur's Gate 3	150.0	NaN
96570	Red Dead Redemption 2	150.0	NaN
65250	HELLDIVERS 2	135.0	NaN
44134	Black Myth: Wukong	130.0	NaN
59122	Grand Theft Auto V Legacy	125.0	NaN
28755	Destiny 2	105.0	NaN
45509	Counter-Strike 2	85.0	NaN
3692	PAYDAY 2	83.0	NaN
85720	Warframe	75.0	NaN
54502	Apex Legends	75.0	NaN

```
[93]: ccu_df = steam_storage_df.dropna(
        subset=["install_size_gb", "Peak CCU"]
    )

plt.figure(figsize=(8,5))

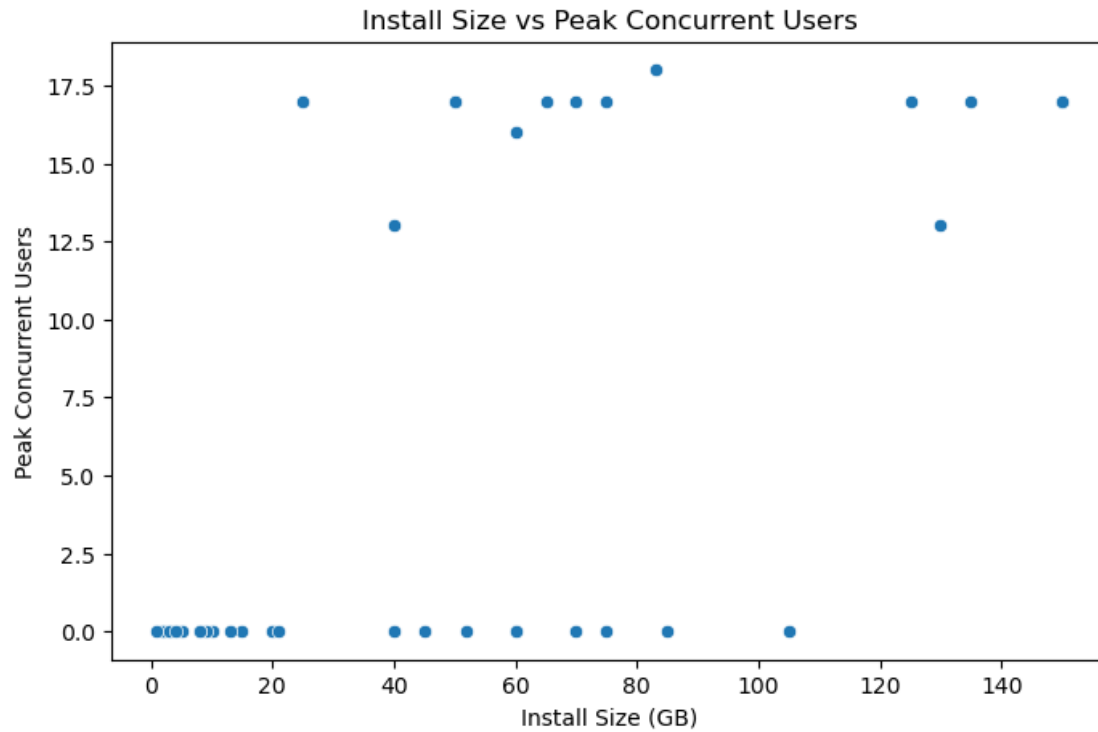
sns.scatterplot(
    data=ccu_df,
    x="install_size_gb",
    y="Peak CCU"
)

plt.title("Install Size vs Peak Concurrent Users")

plt.xlabel("Install Size (GB)")

plt.ylabel("Peak Concurrent Users")

plt.show()
```



```
[94]: rec_df = steam_storage_df.dropna(
        subset=["install_size_gb", "Recommendations"]
    )

    plt.figure(figsize=(8,5))

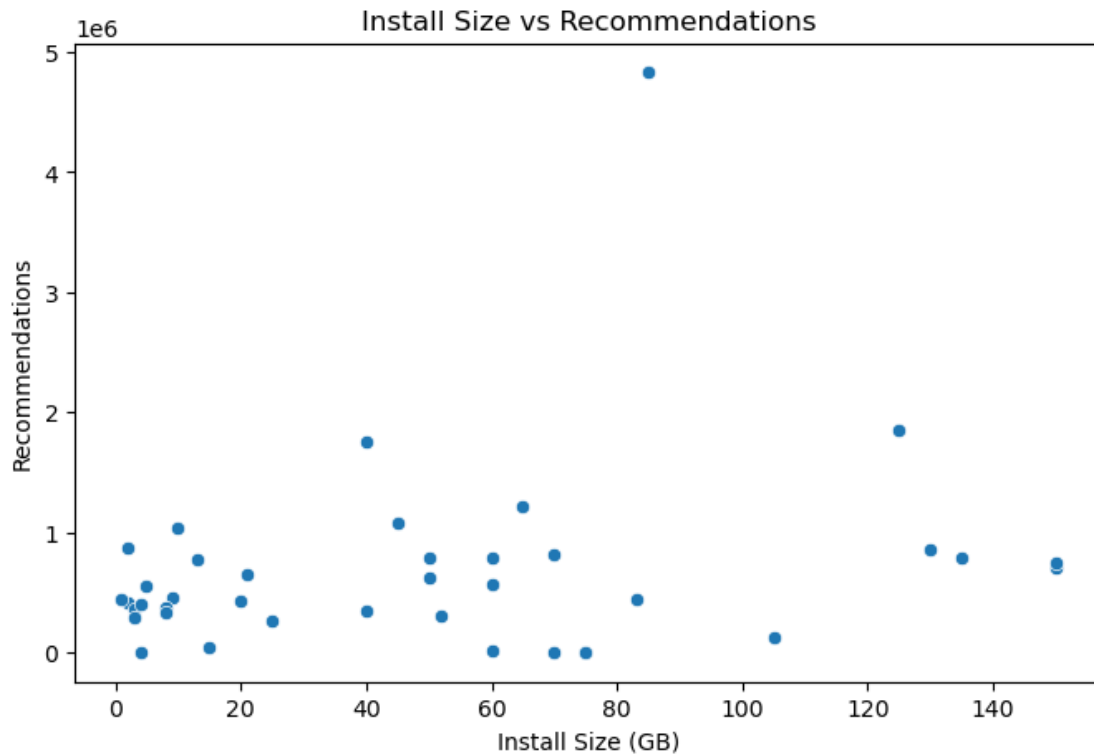
    sns.scatterplot(
        data=rec_df,
        x="install_size_gb",
        y="Recommendations"
    )

    plt.title("Install Size vs Recommendations")

    plt.xlabel("Install Size (GB)")

    plt.ylabel("Recommendations")

    plt.show()
```

```
[100]: meta_df = steam_storage_df[
        (steam_storage_df["install_size_gb"].notna()) &
        (steam_storage_df["Metacritic score"] > 0)
    ]
```

```
[104]: plt.figure(figsize=(8,5))

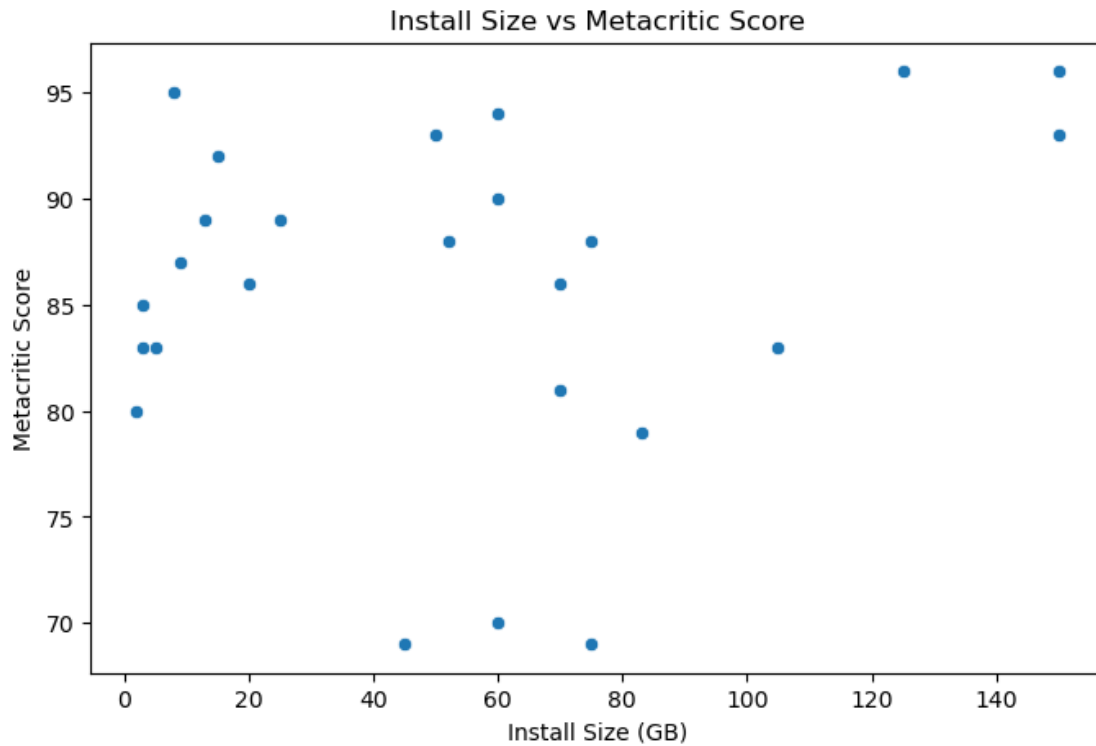
sns.scatterplot(
    data=meta_df,
    x="install_size_gb",
    y="Metacritic score"
)

plt.title("Install Size vs Metacritic Score")

plt.xlabel("Install Size (GB)")

plt.ylabel("Metacritic Score")

plt.show()
```



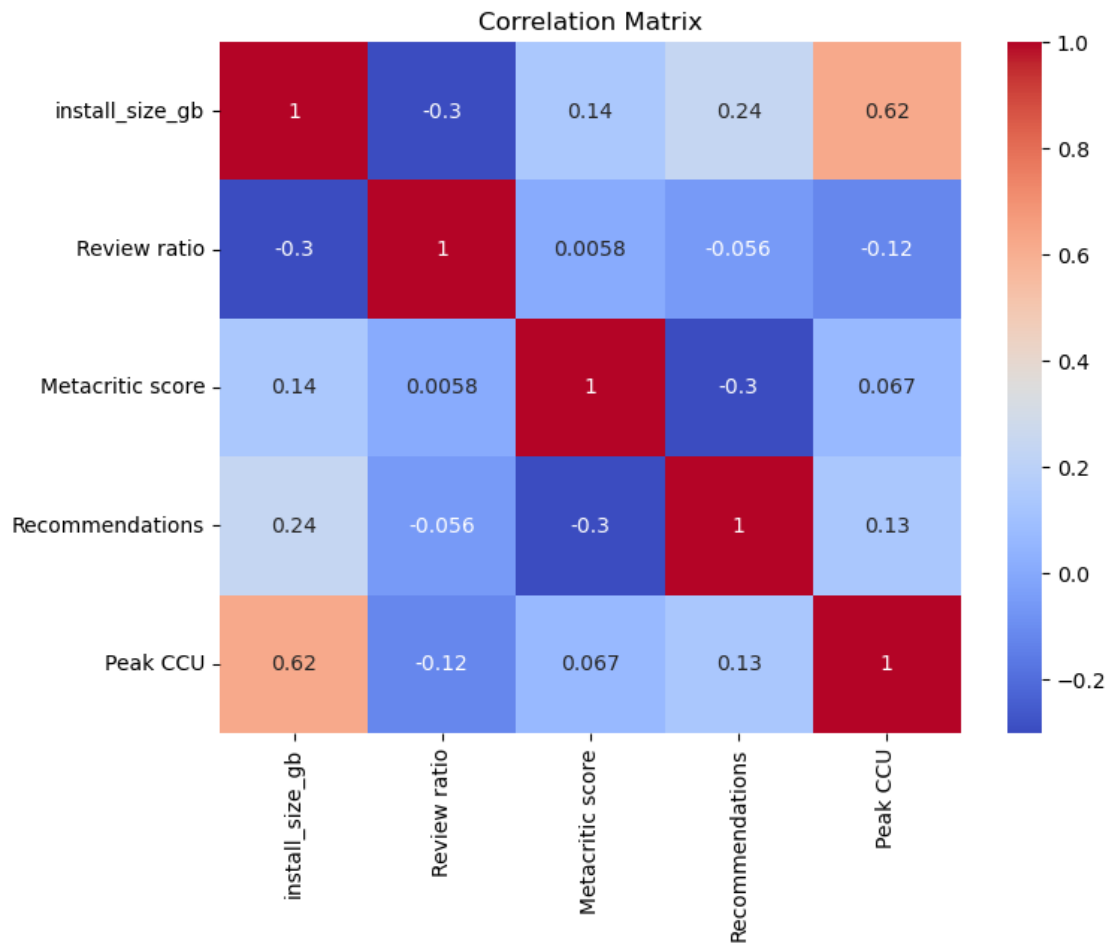
```
[102]: corr_df = steam_storage_df[[
    "install_size_gb",
    "Review ratio",
    "Metacritic score",
    "Recommendations",
    "Peak CCU"
]].dropna()
```

```
[103]: plt.figure(figsize=(8,6))

sns.heatmap(
    corr_df.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```



```
[ ]:
```